

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_1204.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-06 14:27:53

Training Start: 2025-05-06 13-32-50

Training End: 2025-05-06 13-36-47

Training Duration: 3.94 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.5855

MSE: 3.9996

RMSE: 1.9999

R2 Score: 0.7656

Multi-Step Forecast:

Forecast Steps: 20

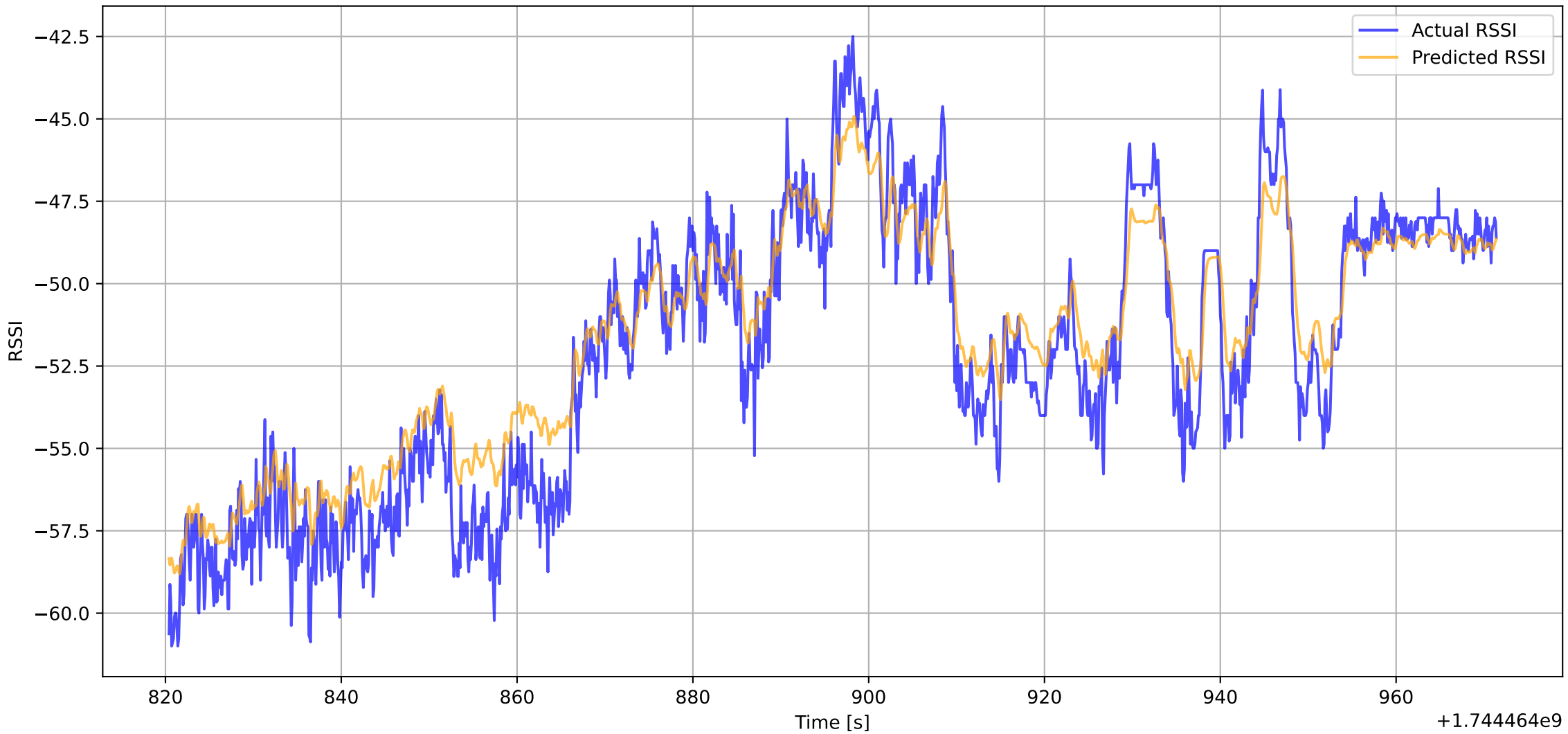
MAE: 0.0432

MSE: 0.0028

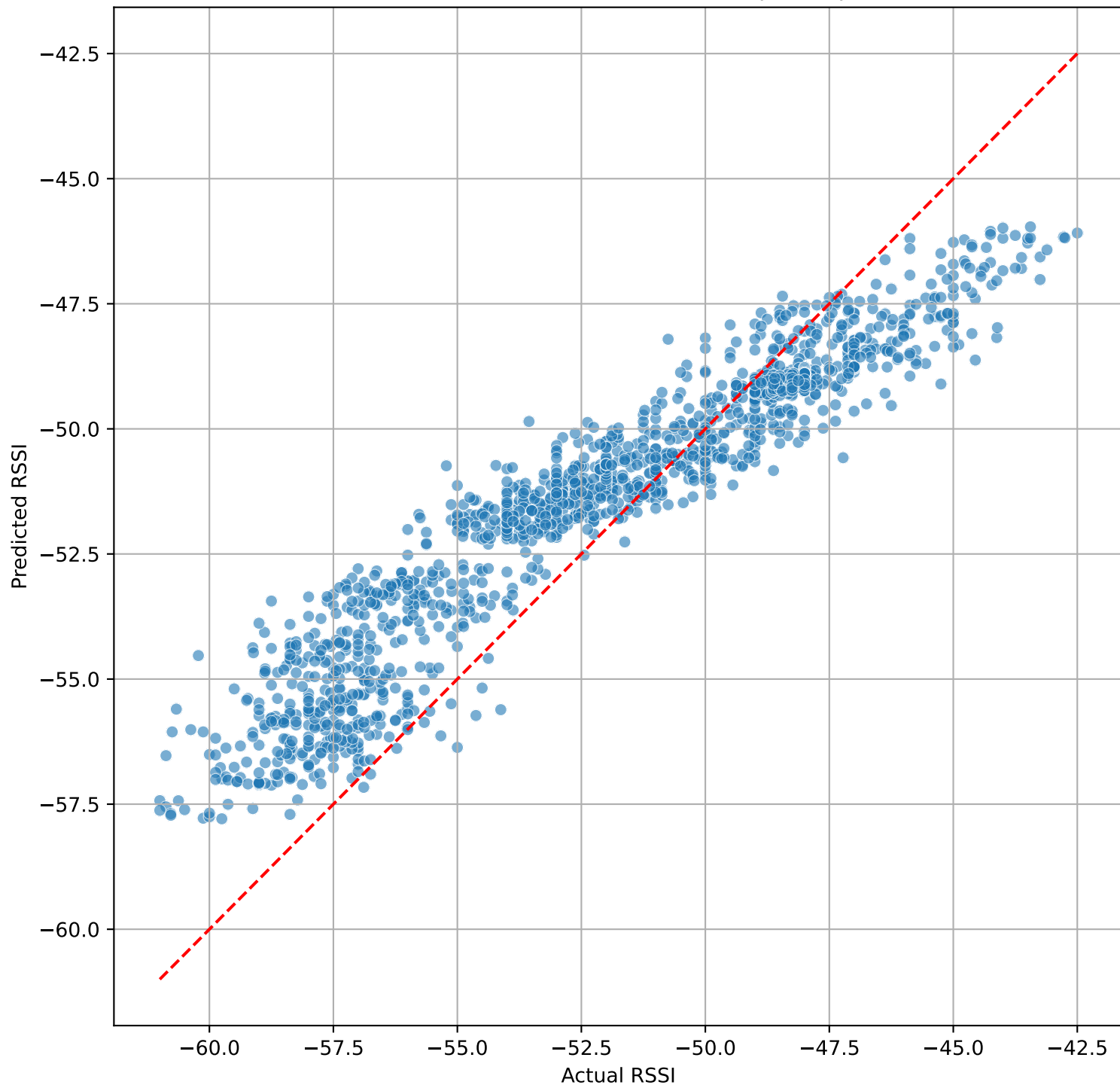
RMSE: 0.0532

R2 Score: 0.8798

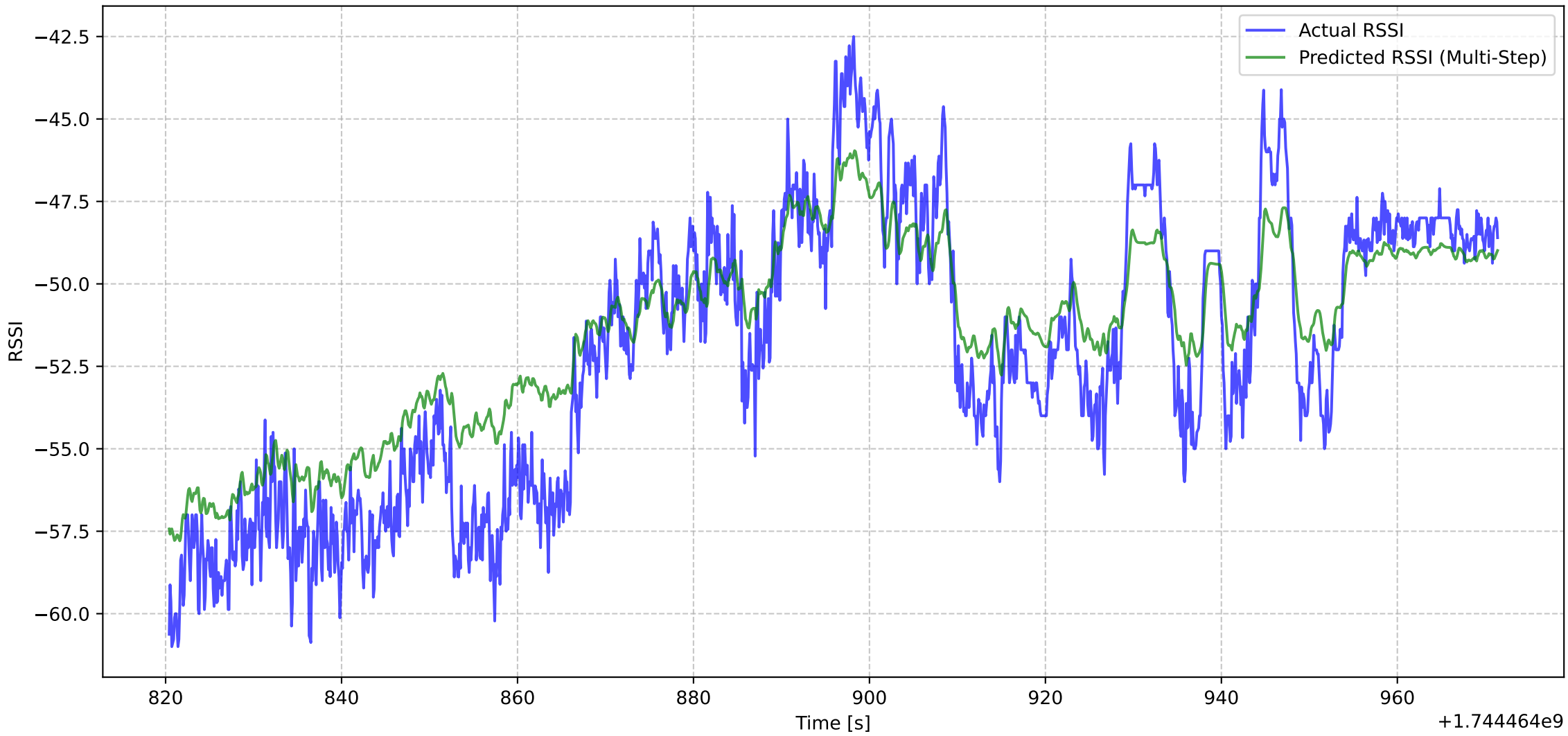
Actual vs Predicted RSSI (First Step)



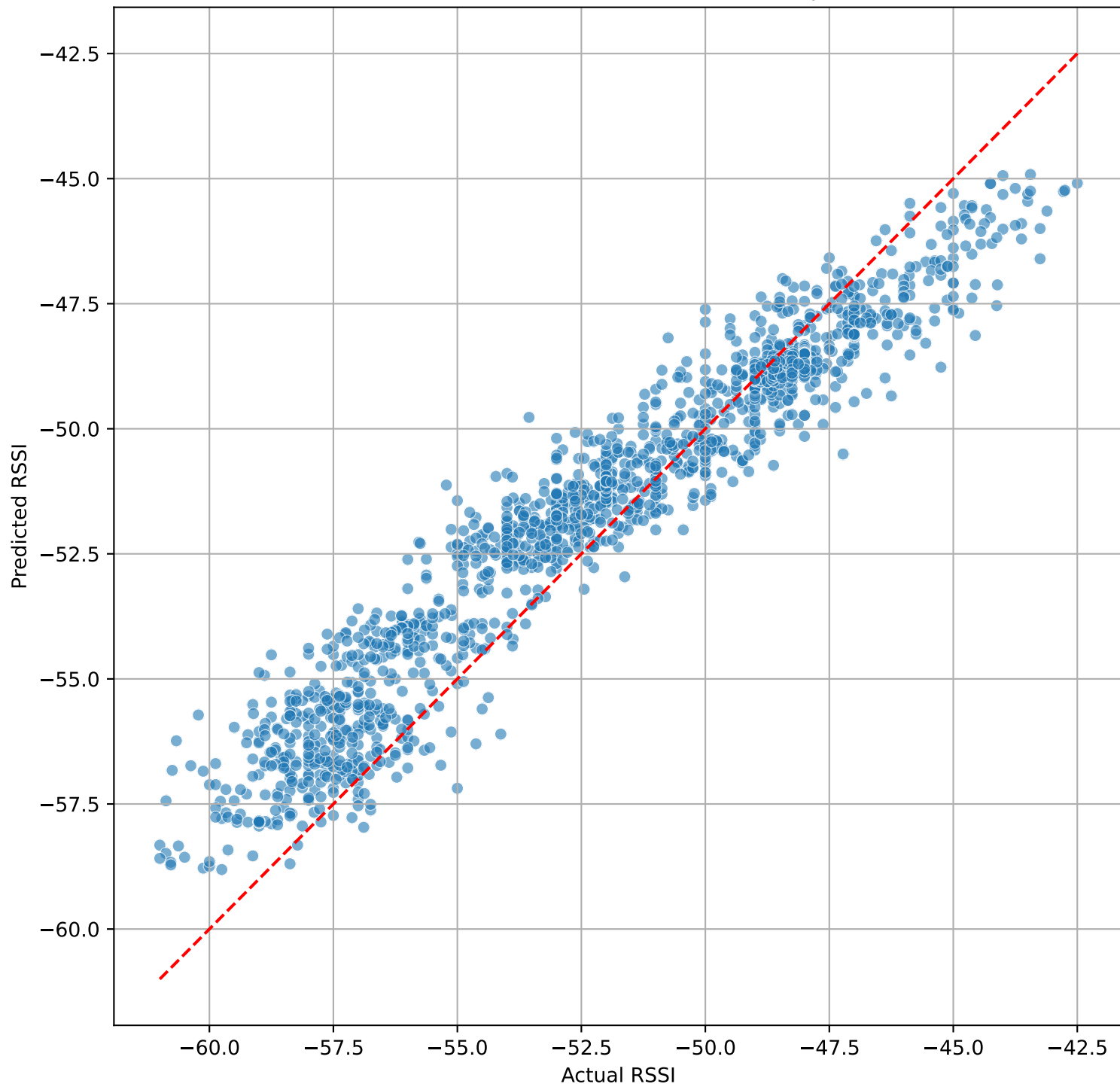
Actual vs Predicted RSSI (Last Step - step 1)



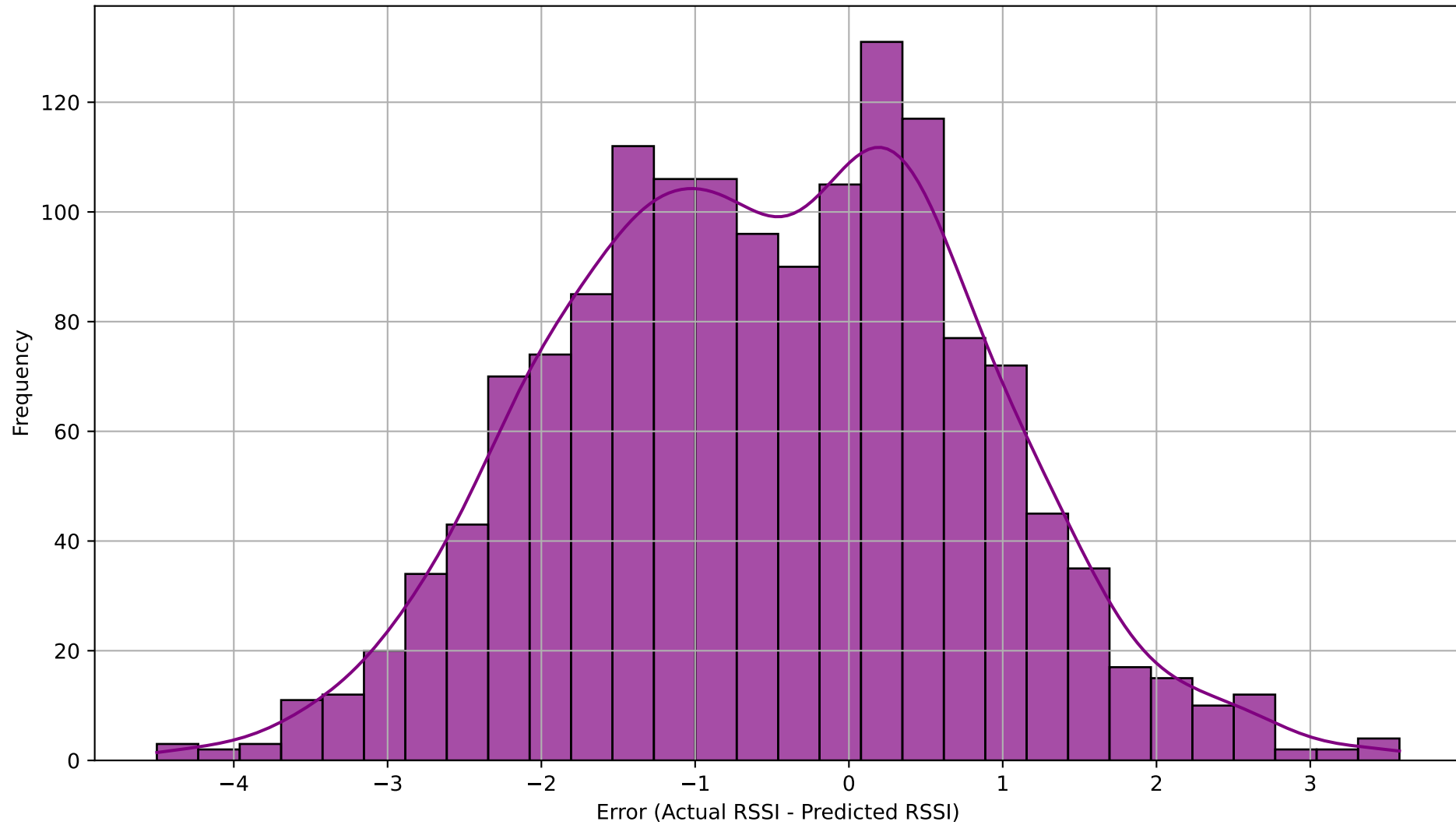
Actual vs Predicted RSSI (Multi-Step)



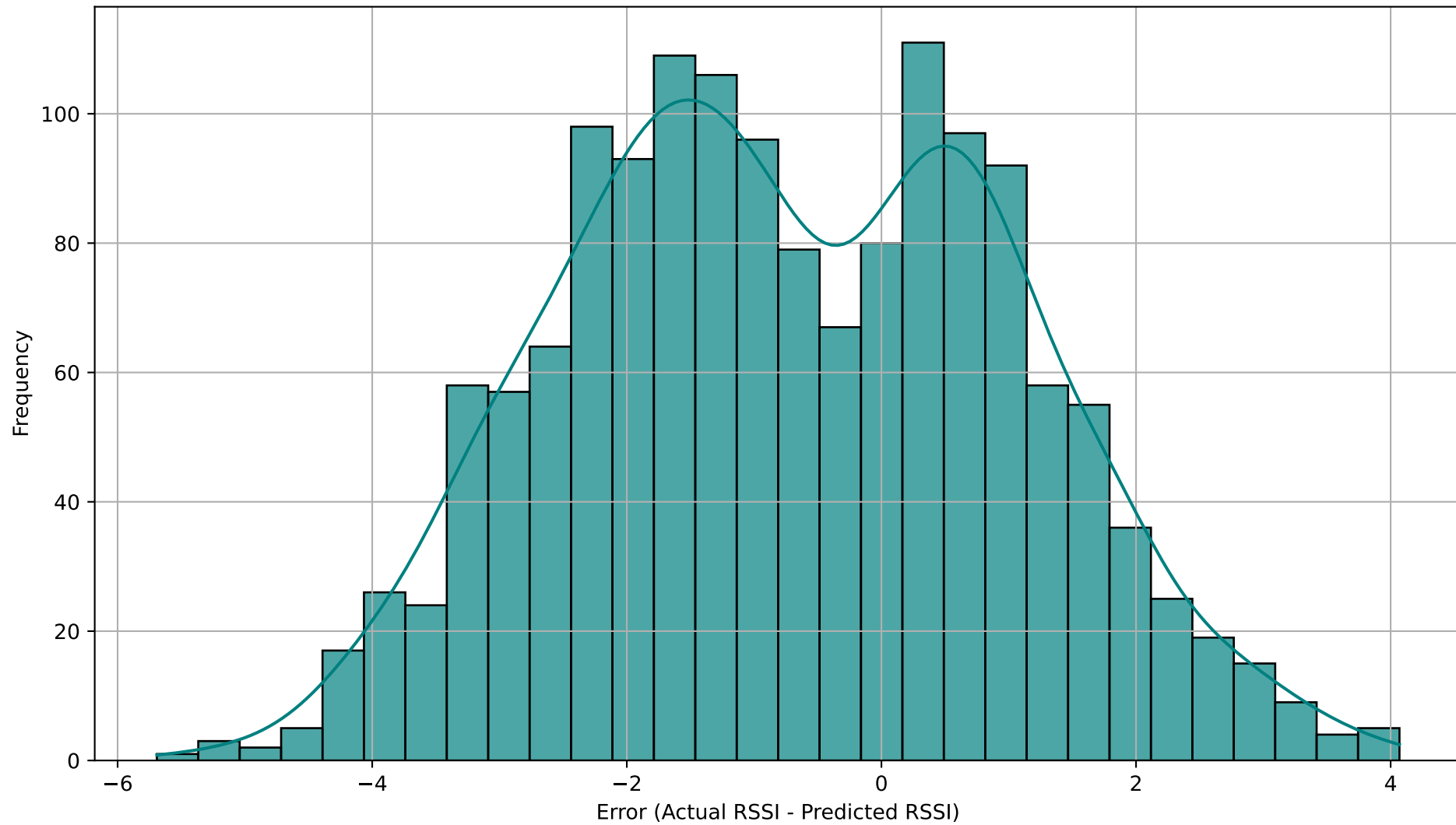
Actual vs Predicted RSSI (First Step)



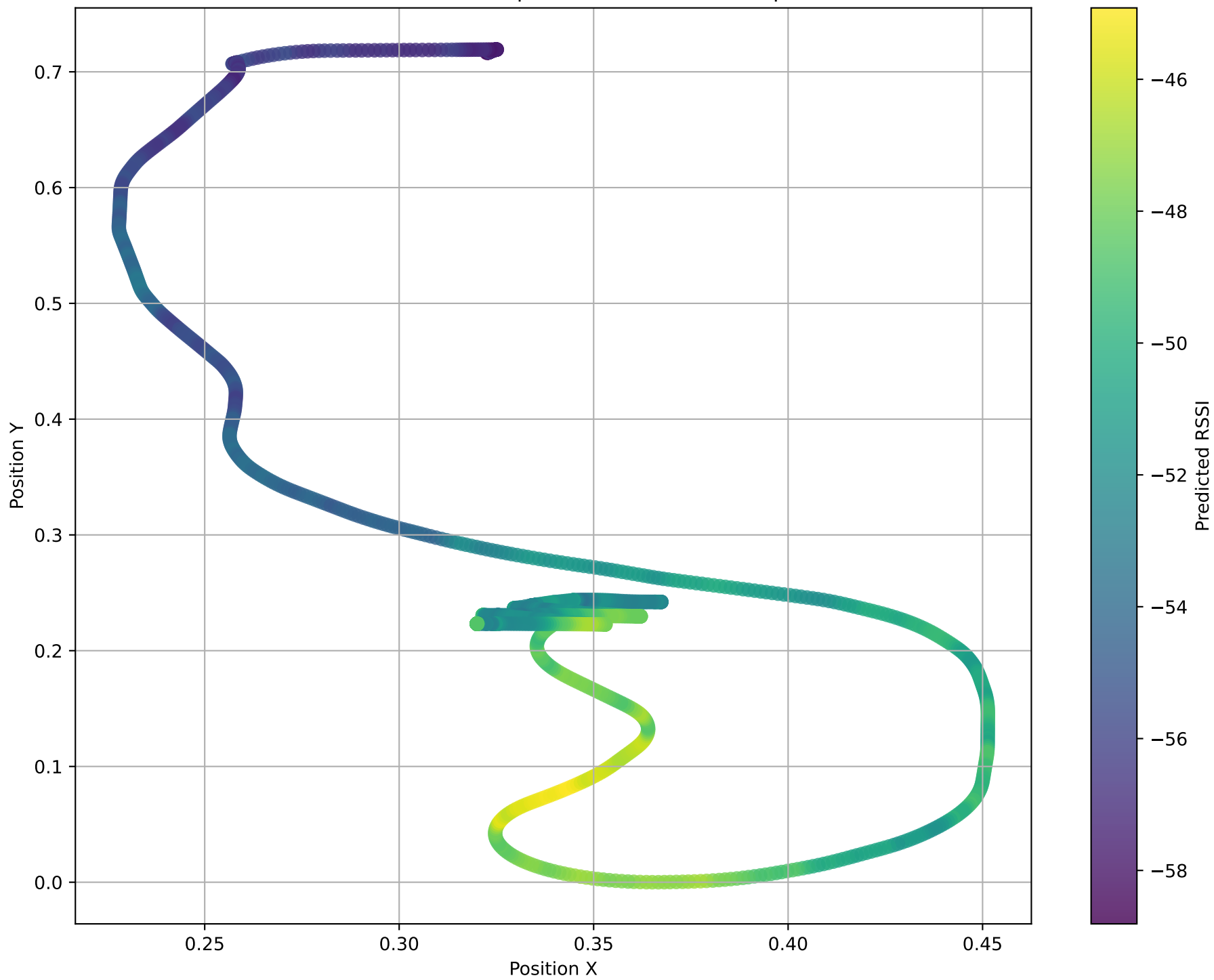
Distribution of Prediction Errors (First Step)



Distribution of Prediction Errors (Last Step - step 1)



Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 1)

